① Matching device → Γ VSWR = $\frac{1+|\Gamma|}{1-|\Gamma|}$ RL = Γ^2
→ Bandwidth RL = $-20\log|\Gamma|$ ✓

② Directional device
→ Pola radiasi

Fundamental Parameters of Antennas

2.2 RADIATION PATTERN

An antenna *radiation pattern* or *antenna pattern* is defined as "a mathematical function or a graphical representation of the radiation properties of the antenna as a function of space coordinates. In most cases, the radiation pattern is determined in the far-field region and is represented as a function of the directional coordinates.

Radiation properties include power flux density, radiation intensity, field strength, directivity, phase or polarization."

- Kartesian A(x, y, z)
- Silinder A(ρ, ϕ, z)
- Bola A(r, θ, ϕ)

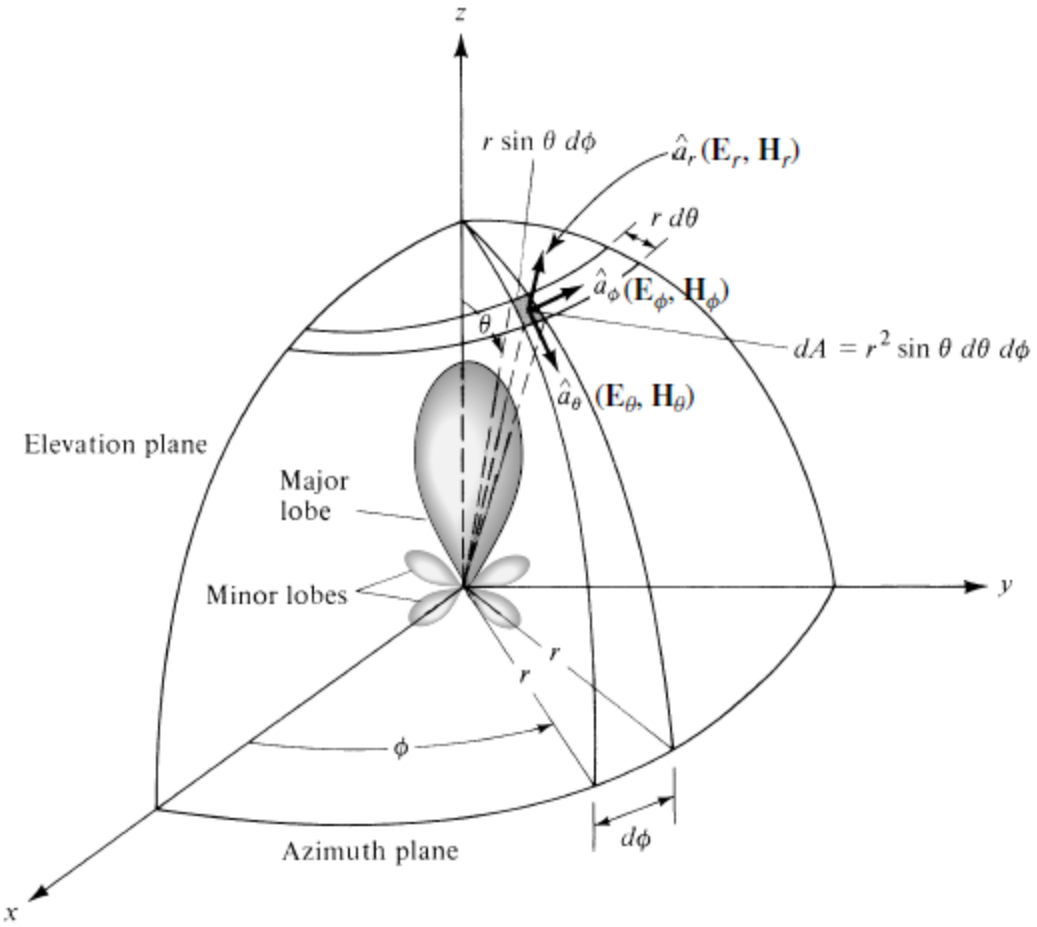
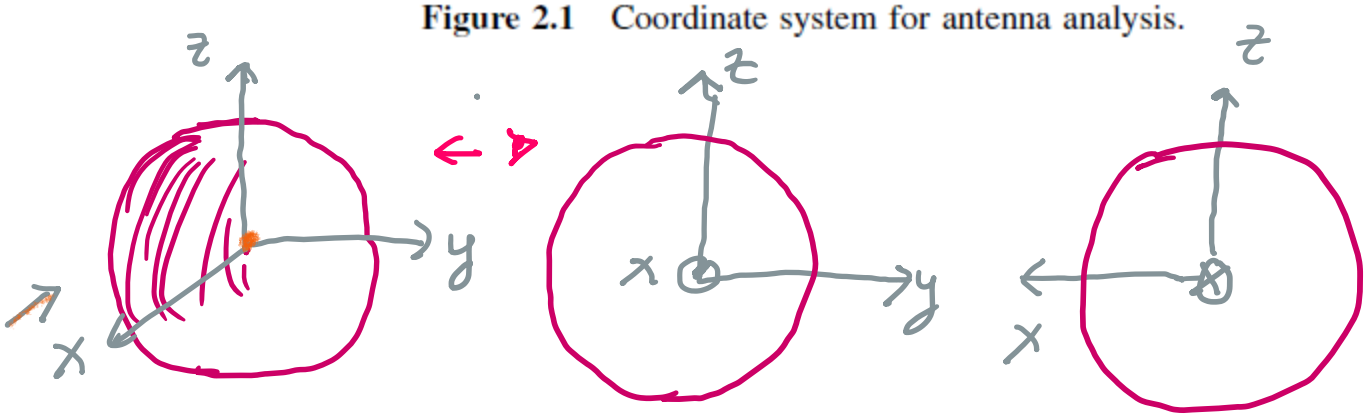
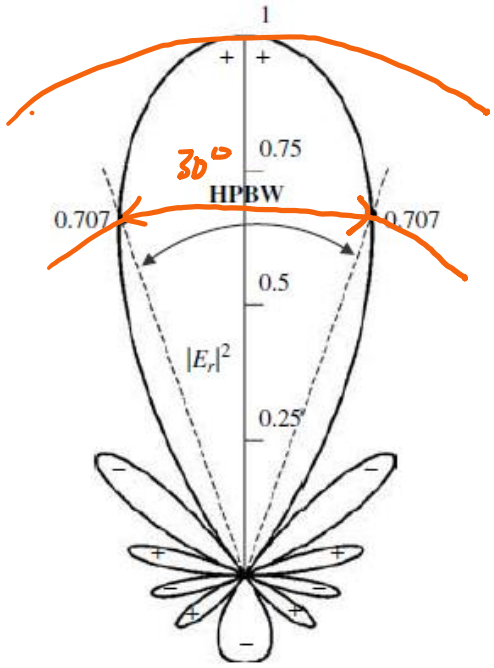
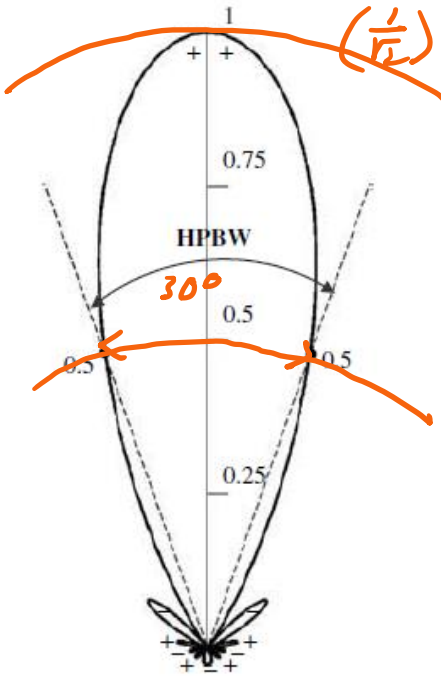


Figure 2.1 Coordinate system for antenna analysis.

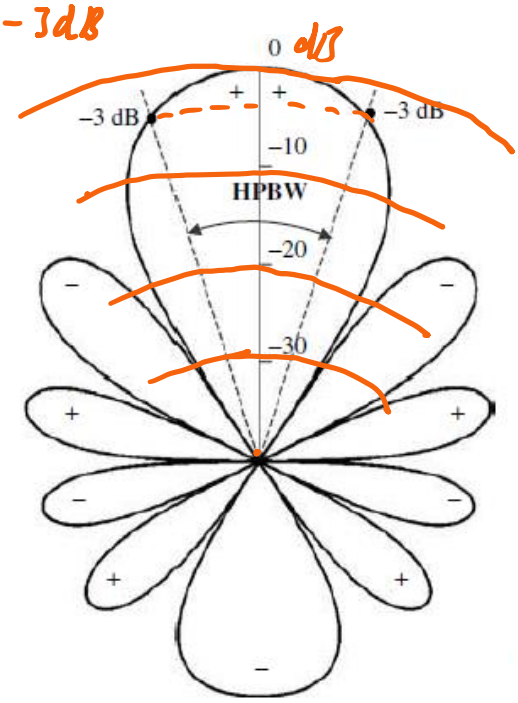




(a) Field pattern (in linear scale)



(b) Power pattern (in linear scale)



(c) Power pattern (in dB)

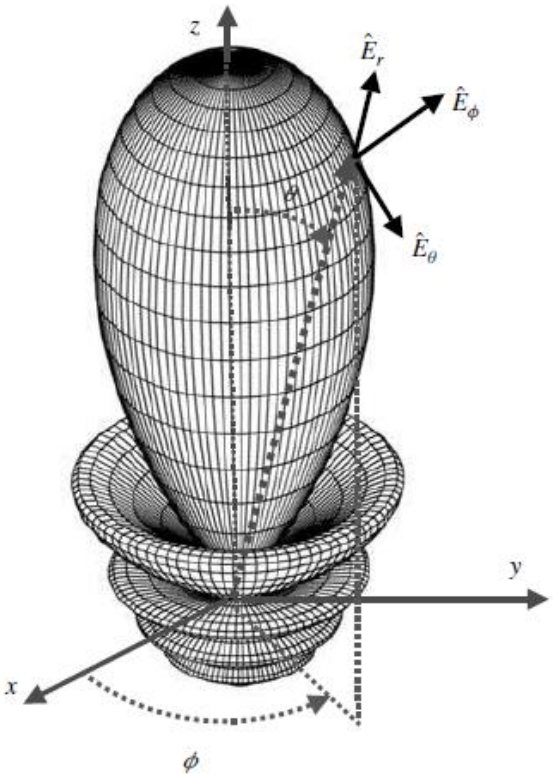
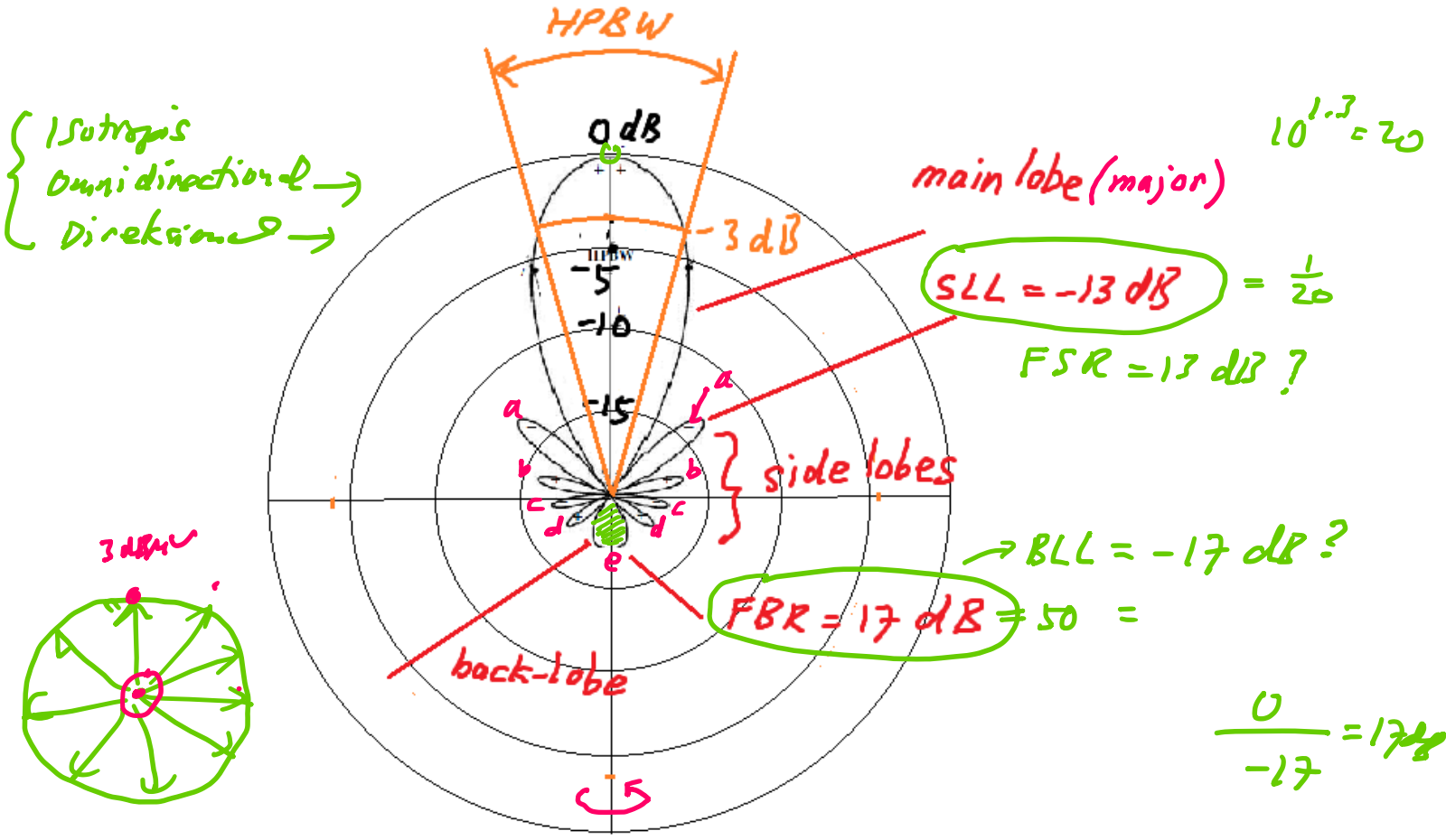


Figure 2.2 Two-dimensional normalized *field* pattern (*linear scale*), *power* pattern (*linear scale*), and *power* pattern (*in dB*) of a 10-element linear array with a spacing of $d = 0.25\lambda$.

2.2.1 Radiation Pattern Lobes



2.2.2 Isotropic, Directional, and Omnidirectional Patterns

An *isotropic* radiator is defined as "a hypothetical lossless antenna having equal radiation in all directions."

Although it is ideal and not physically realizable, it is often taken as a reference for expressing the directive properties of actual antennas.

A *directional* antenna is one "having the property of radiating or receiving electromagnetic waves more effectively in some directions than in others.

omnidirectional, and it is defined as one "having an essentially nondirectional pattern in a given plane (in this case in azimuth) and a directional pattern in any orthogonal plane (in this case in elevation)." An *omnidirectional* pattern is then a special type of a *directional* pattern.

2.2.3 Principal Patterns

The *E-plane* is defined as "the plane containing the electric-field vector and the direction of maximum radiation," and the *H-plane* as "the plane containing the magnetic-field vector and the direction of maximum radiation."

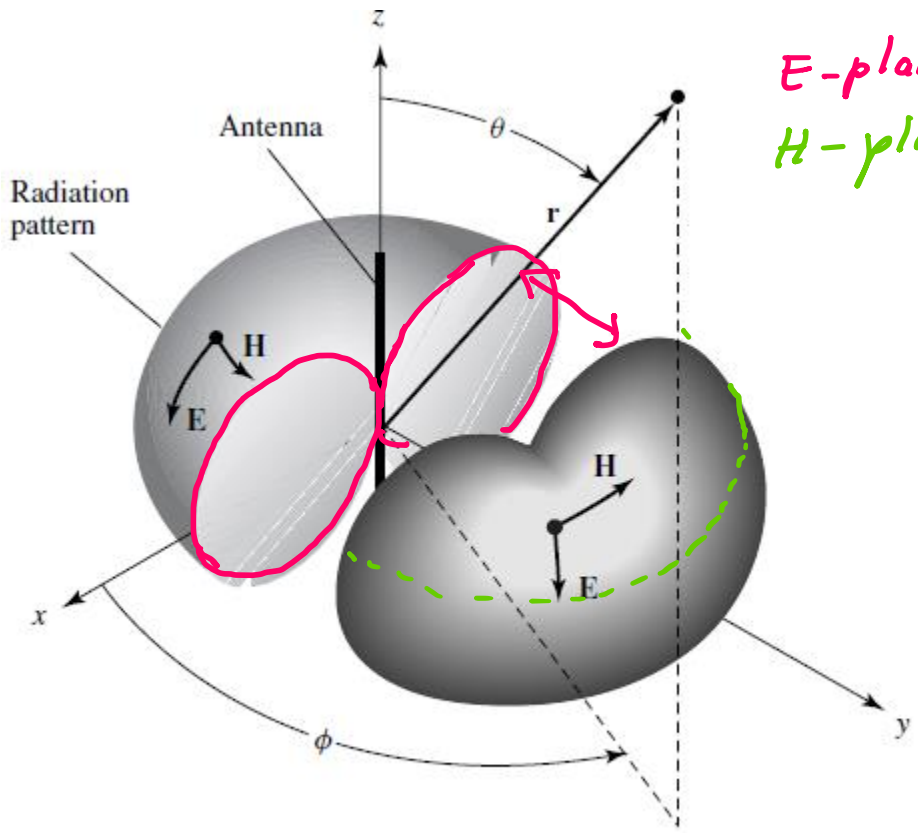
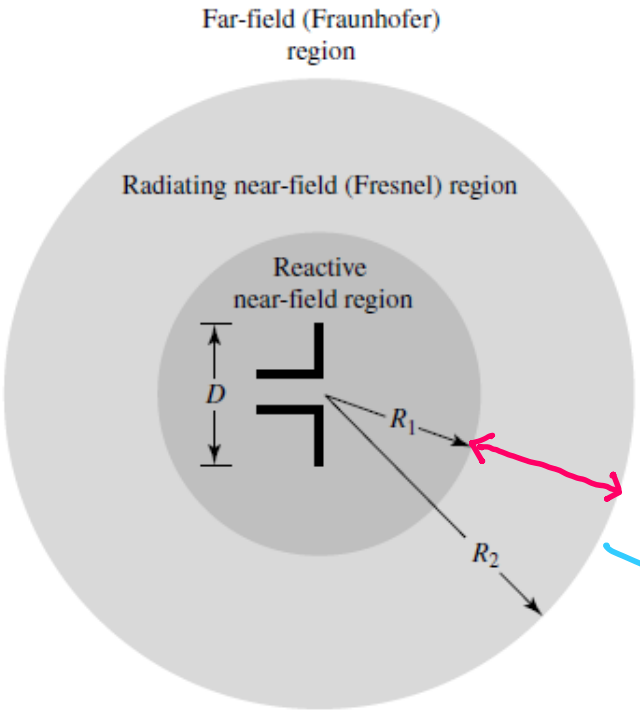


Figure 2.6 Omnidirectional antenna pattern.

2.2.4 Field Regions



$R_1 = 0.62\sqrt{D^3/\lambda}$ medan dekat reaktif

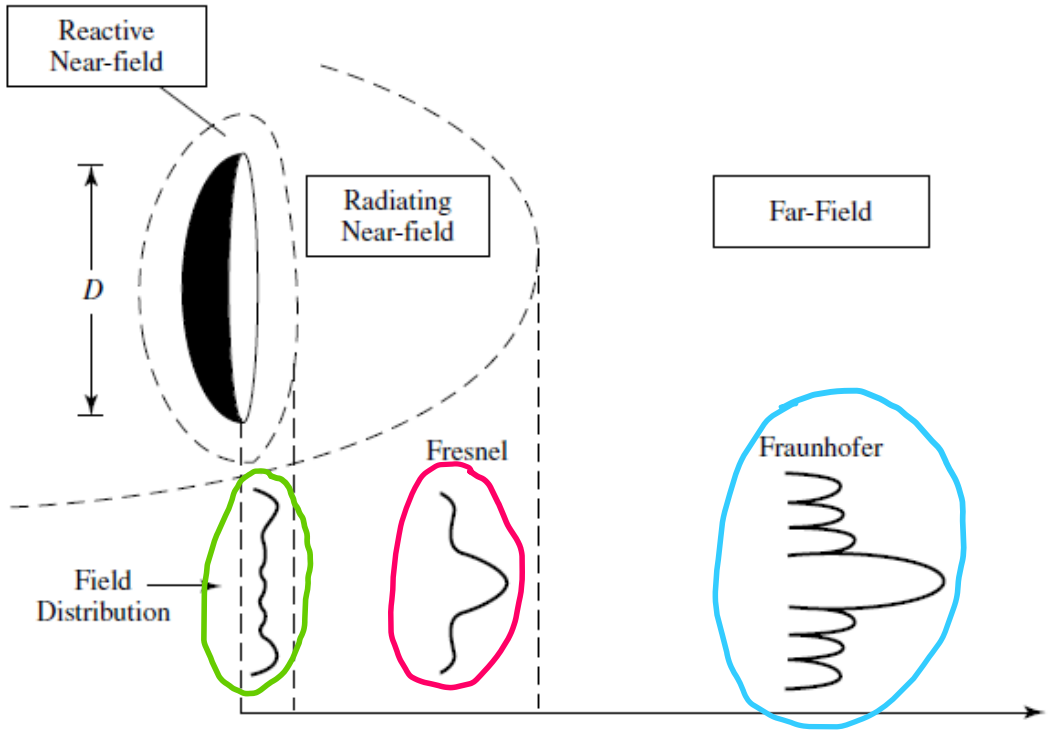
$R_2 = 2D^2/\lambda$ medan dekat radiasi

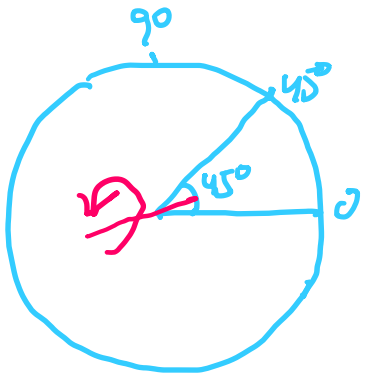
$> R_2$ medan jauh

$R_1 = 0.62\sqrt{D^3/\lambda}$

$R_2 = 2D^2/\lambda$

Figure 2.7 Field regions of an antenna.





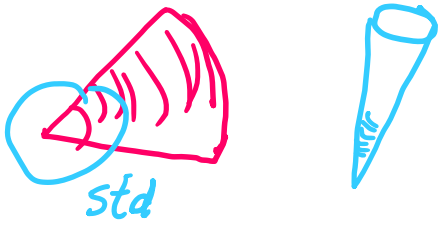
$360° = 2\pi$ rad

$180° = \pi$ rad

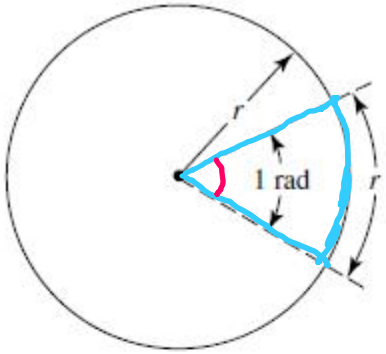
$45° = \frac{\pi}{4}$ rad

$1 \text{ rad} = \frac{180°}{3.14}$

$\approx 57,3°$



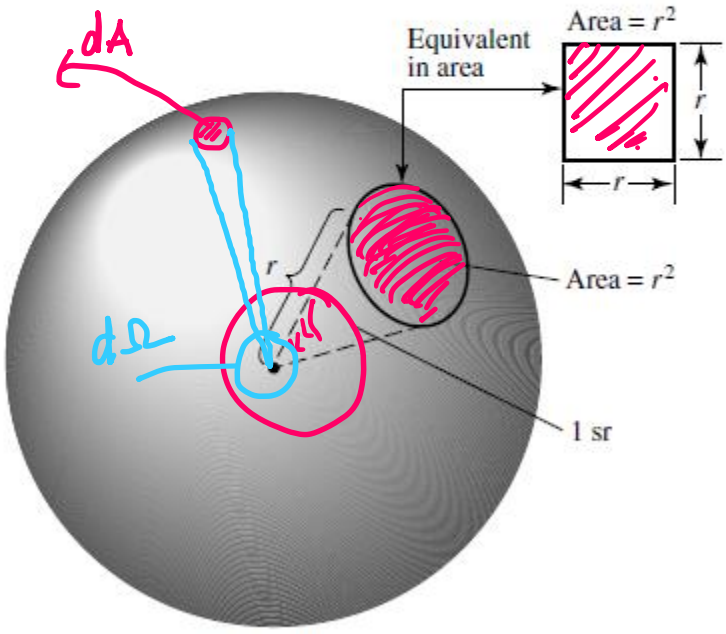
2.2.5 Radian and Steradian



(a) Radian

$dA = r^2 \sin\theta \, d\theta \, d\phi$

$d\Omega = \sin\theta \, d\theta \, d\phi$



(b) Steradian

Figure 2.10 Geometrical arrangements for defining a radian and a steradian.

$$dA = r^2 \sin\theta \, d\theta \, d\phi \quad (\text{m}^2) \tag{2-1}$$

$$d\Omega = \frac{dA}{r^2} = \sin\theta \, d\theta \, d\phi \quad (\text{sr}) \tag{2-2}$$

Example 2.1

For a sphere of radius r, find the solid angle Ω_A (in square radians or steradians) of a spherical cap on the surface of the sphere over the north-pole region defined by spherical angles of $0 \le \theta \le 30°$, $0 \le \phi \le 180°$. Refer to Figures 2.1 and 2.10. Do this

a. exactly.
b. using $\Omega_A \approx \Delta\Theta_1 \cdot \Delta\Theta_2$, where $\Delta\Theta_1$ and $\Delta\Theta_2$ are two perpendicular angular separations of the spherical cap passing through the north pole.

Compare the two.

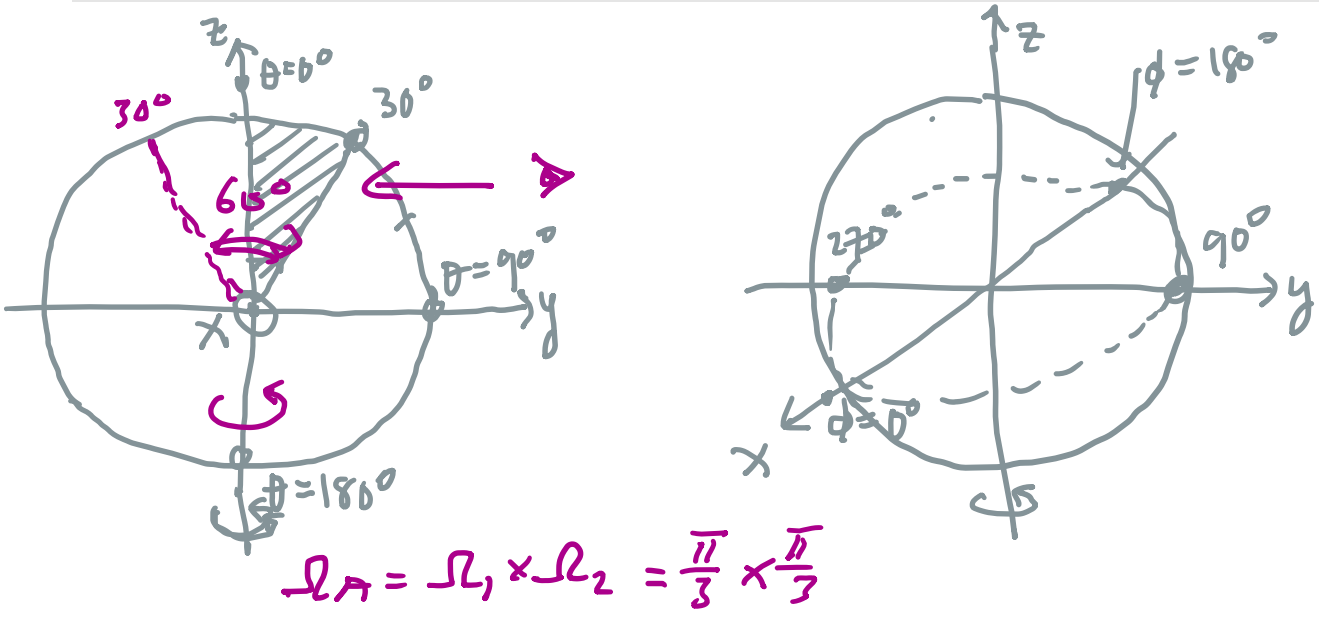
a. Using (2-2), we can write that

$$\Omega_A = \int_0^{360°} \int_0^{30°} d\Omega = \int_0^{2\pi} \int_0^{\pi/6} \sin\theta \, d\theta \, d\phi = \int_0^{2\pi} d\phi \int_0^{\pi/6} \sin\theta \, d\theta$$

$$= 2\pi[-\cos\theta]_0^{\pi/6} = 2\pi[-0.867 + 1] = 2\pi(0.133) = 0.83566$$

b. $\Omega_A \approx \Delta\Theta_1 \cdot \Delta\Theta_2 \overbrace{=}^{\Delta\Theta_1 = \Delta\Theta_2} (\Delta\Theta_1)^2 = \frac{\pi}{3}\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9} = 1.09662$

It is apparent that the approximate beam solid angle is about 31.23% in error.



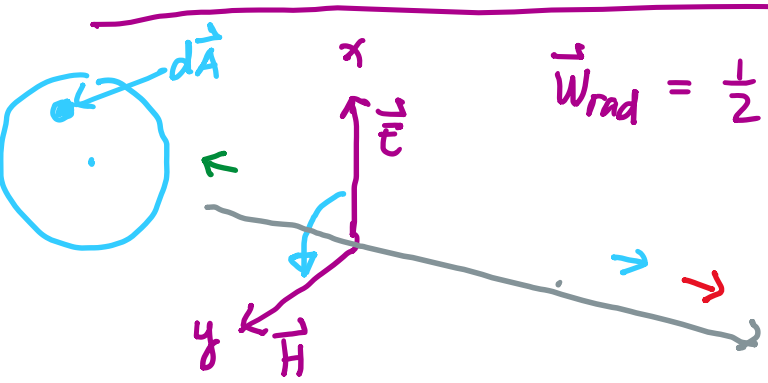
$$\Omega_A = \Omega_1 \times \Omega_2 = \frac{\pi}{3} \times \frac{\pi}{3}$$

ⓐ $\Omega_A = \int_{\phi=0}^{360} \int_{\theta=0}^{30} \sin\theta \, d\theta \, d\phi = \int_{\phi=0}^{360^\circ} d\phi \int_{\theta=0}^{30^\circ} \sin\theta \, d\theta = \phi\Big|_0^{2\pi} \left(-\cos\theta\Big|_0^{30^\circ}\right)$

$= -2\pi \cos\theta\Big|_0^{30^\circ} = -2\pi(\cos 30^\circ - 1) = -2\pi(0.87 - 1) = 0.82$ std

ⓑ $\Omega_A = \Omega_1 \times \Omega_2 = \left(\frac{\pi}{3}\right)\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9} = 1.1$ std

---



$\vec{W}_{rad} = \frac{1}{2} Re[\vec{E} \times \vec{H}^*] = \frac{1}{2} |\vec{E}||\vec{H}|$

$\frac{W}{m^2} \Rightarrow P_r =$

2.3 RADIATION POWER DENSITY

Power density = rapat daya (W/m^2)

The quantity used to describe the power associated with an electromagnetic wave is the instantaneous Poynting vector defined as

$$\mathcal{W} = \mathcal{E} \times \mathcal{H} \tag{2-3}$$

$\mathcal{W}$ = instantaneous Poynting vector (W/m^2)
$\mathcal{E}$ = instantaneous electric-field intensity (V/m)
$\mathcal{H}$ = instantaneous magnetic-field intensity (A/m)

$$\mathcal{P} = \oint_S \mathcal{W} \cdot ds = \oint_S \mathcal{W} \cdot \hat{\mathbf{n}} \, da \tag{2-4}$$

$\mathcal{P}$ = instantaneous total power (W)
$\hat{\mathbf{n}}$ = unit vector normal to the surface
da = infinitesimal area of the closed surface (m^2)

$$\boxed{\mathbf{W}_{av}(x, y, z) = [\mathcal{W}(x, y, z; t)]_{av} = \tfrac{1}{2}\text{Re}[\mathbf{E} \times \mathbf{H}^*]} \quad (\text{W/m}^2) \tag{2-8}$$

$$P_{\text{rad}} = P_{\text{av}} = \oint_S \mathbf{W}_{\text{rad}} \cdot d\mathbf{s} = \oint_S \mathbf{W}_{\text{av}} \cdot \hat{\mathbf{n}}\, da$$
$$= \frac{1}{2} \oint_S \text{Re}(\mathbf{E} \times \mathbf{H}^*) \cdot d\mathbf{s} \quad \checkmark \qquad (2\text{-}9)$$

Example 2.2

The radial component of the radiated power density of an antenna is given by

$$\mathbf{W}_{\text{rad}} = \hat{\mathbf{a}}_r W_r = \hat{\mathbf{a}}_r A_0 \frac{\sin\theta}{r^2} \ (\text{W/m}^2)$$

where A_0 is the peak value of the power density, θ is the usual spherical coordinate, and $\hat{\mathbf{a}}_r$ is the radial unit vector. Determine the total radiated power.

Solution: For a closed surface, a sphere of radius r is chosen. To find the total radiated power, the radial component of the power density is integrated over its surface. Thus

$$P_{\text{rad}} = \oint_S \mathbf{W}_{\text{rad}} \cdot \hat{\mathbf{n}}\, da$$
$$= \int_0^{2\pi} \int_0^{\pi} \left(\hat{\mathbf{a}}_r A_0 \frac{\sin\theta}{r^2}\right) \cdot (\hat{\mathbf{a}}_r r^2 \sin\theta\, d\theta\, d\phi) = \pi^2 A_0 \ (\text{W}) \quad \checkmark$$

$\hat{a}_r \cdot \hat{a}_r = 1$

$$P_{\text{rad}} = A_0 \int_0^{2\pi} \int_0^{\pi} \sin^2\theta\, d\theta\, d\phi = A_0 \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin^2\theta\, d\theta = 2\pi A_0\, G$$

$$\cos 2\theta = 1 - 2\sin^2\theta \rightarrow 2\sin^2\theta = 1 - \cos 2\theta \rightarrow \sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$$

$$G = \int_{\theta=0}^{\pi} \frac{1}{2}\, d\theta - \int_{\theta=0}^{\pi} \frac{1}{2}\cos 2\theta\, d\theta = \frac{\pi}{2} - \frac{1}{2}\cdot\frac{1}{2}\int_{\theta=0}^{\pi} \cos 2\theta\, d2\theta = \frac{\pi}{2} - \frac{1}{4}\sin 2\theta \Big|_0^{\pi}$$

$$= \frac{\pi}{2} - \frac{1}{4}[0-0] = \frac{\pi}{2}$$

$$P_{\text{rad}} = 2\pi A_0 \times \frac{\pi}{2} = \pi^2 A_0 \quad \checkmark$$

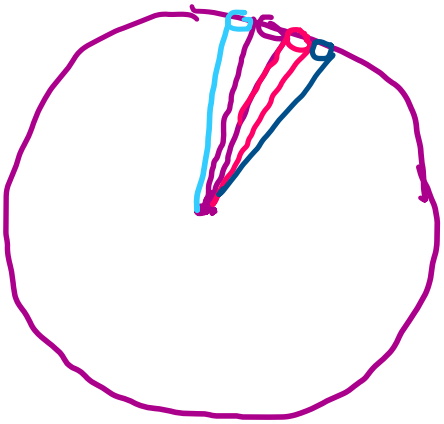
An isotropic radiator is an ideal source that radiates equally in all directions.

$$P_{\text{rad}} = \oint_S \mathbf{W}_0 \cdot d\mathbf{s} = \int_0^{2\pi} \int_0^{\pi} [\hat{\mathbf{a}}_r W_0(r)] \cdot [\hat{\mathbf{a}}_r r^2 \sin\theta \, d\theta \, d\phi] = 4\pi r^2 W_0 \qquad (2\text{-}10)$$

$$\mathbf{W}_0 = \hat{\mathbf{a}}_r W_0 = \hat{\mathbf{a}}_r \left(\frac{P_{\text{rad}}}{4\pi r^2} \right) \quad (\text{W/m}^2) \qquad (2\text{-}11)$$

$W_{rad} = \frac{W}{m^2} \rightarrow U = r^2 W_{rad} = \frac{W}{std}$

$d\Omega$



2.4 RADIATION INTENSITY

Radiation intensity in a given direction is defined as "the power radiated from an antenna per unit solid angle."

$$\boxed{U = r^2 W_{\text{rad}}} \qquad \left(\frac{W}{std}\right) \qquad (2\text{-}12)$$

$$P_{\text{rad}} = \oint_{\Omega} U \, d\Omega = \int_0^{2\pi} \int_0^{\pi} U \sin\theta \, d\theta \, d\phi \qquad (2\text{-}13)$$

$d\Omega$

$d\Omega$ = element of solid angle = $\sin\theta \, d\theta \, d\phi$.

Example 2.3

For the problem of Example 2.2, find the total radiated power using (2-13).

Solution: Using (2-12)

$$U = r^2 W_{\text{rad}} = A_0 \sin\theta$$

and by (2-13)

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi} U \sin\theta \, d\theta \, d\phi = A_0 \int_0^{2\pi} \int_0^{\pi} \sin^2\theta \, d\theta \, d\phi = \pi^2 A_0$$

For an isotropic source U will be independent of the angles θ and ϕ, as was the case for W_{rad}.

$$P_{\text{rad}} = \oint_{\Omega} U_0 \, d\Omega = U_0 \oint_{\Omega} d\Omega = 4\pi U_0 \tag{2-14}$$

$$U_0 = \frac{P_{\text{rad}}}{4\pi} \tag{2-15}$$

2.6 DIRECTIVITY

$$D = \frac{U}{U_0} = \frac{4\pi U}{P_{\text{rad}}} \tag{2-16}$$

$$D_{\max} = D_0 = \frac{U|_{\max}}{U_0} = \frac{U_{\max}}{U_0} = \frac{4\pi U_{\max}}{P_{\text{rad}}} \tag{2-16a}$$

D = directivity (dimensionless)
D_0 = maximum directivity (dimensionless)

U = radiation intensity (W/unit solid angle)
$U_{\max}$ = maximum radiation intensity (W/unit solid angle)
U_0 = radiation intensity of isotropic source (W/unit solid angle)
P_{rad} = total radiated power (W)

Example 2.5

As an illustration, find the maximum directivity of the antenna whose radiation intensity is that of Example 2.2. Write an expression for the directivity as a function of the directional angles θ and ϕ.

$$U = r^2 W_{\text{rad}} = A_0 \sin\theta$$

The maximum radiation is directed along $\theta = \pi/2$. Thus

$$U_{\max} = A_0$$

In Example 2.2 it was found that

$$P_{\text{rad}} = \pi^2 A_0$$

$$D_0 = \frac{4\pi U_{\max}}{P_{\text{rad}}} = \frac{4}{\pi} = 1.27$$

$$D = D_0 \sin\theta = 1.27 \sin\theta$$